Affine STARK transition: technical proof composition

Pinned review: 2026-11-08. **Producer source:** `CLAIMS.md` and Lean module `formal/CryptoResearch/Stark/AffineTransition.lean` (SHA-256 `3311451f0dfa19c9c696b872bb3b85d0ed60e68a75c89c871052cbd39dadfa0d`).

Scoped result: Seven ring-expression propositions are supported by passing Lean machine evidence, independent blind derivation, and independent source/type review. The native reviewer confirmed seven built declarations, their types and axioms, and two nontriviality controls. The complete technical outcome remains `inconclusive` pending independent closure of J3′s trust boundary. The Rust implementation and STARK protocol bridges remain open.

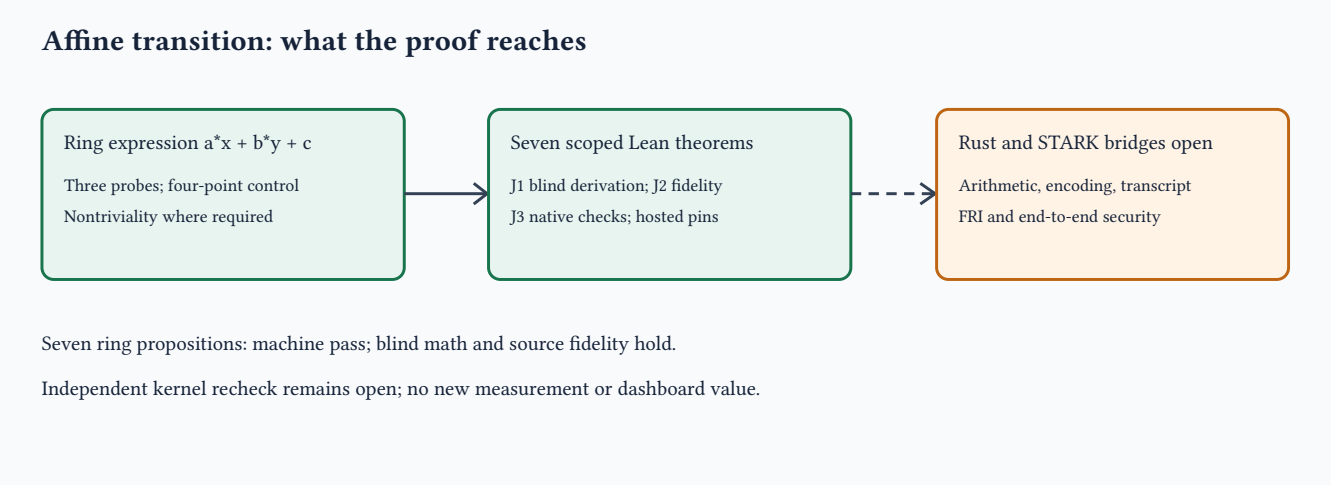

Figure 1: Solid arrow: the reviewed ring-expression proof. Dashed arrow: unproved implementation and security bridges.

Review and machine evidence

The precommitted plan assigned J1 blind algebra, J2 source/type fidelity, and J3 machine-boundary review to three separate reviewers. The normalized independence check passed for all three reports and attestations. J1 and J2 hold all seven numbered claims. J3 holds the type, axiom, and boundary-control subchecks; its full-boundary verdict was inconclusive because the older run did not prove installed dependency cleanliness or supply an independent kernel recheck.

Hosted run 41109712394 built the seven-declaration module and retained types, axiom prints and zero-ring rejection controls. Hosted run 42423957020 later built the same theorem bytes within the full workspace and gave a `machine_verified` receipt for the registered `three_probe_extensionality` target. Its identical pre/post dependency-pin receipts report clean installed revisions. An artifact-wide audit matched formal and verifier input bytes to crypto-autoresearcher main `4239648a889fe97d976a99862d178ba50324dc6e`. No independent `lean4checker` or `nanoda` result is claimed.

Claim map

1. `recover_coefficients`: three evaluations recover the tuple `(a,b,c)`.
2. `coefficients_eq_of_probes`: equal values at the three probes force equal coefficients.
3. `three_probe_extensionality`: equal probes force equal values at all pairs.
4. `multiplication_not_affine`: in a nontrivial ring, multiplication is not globally affine.
5. `two_probes_insufficient`: omitting the third probe admits distinct transitions.
6. `zero_ring_product_control`: the zero ring is the required exception.
7. `multiplication_not_affine_on_four_probes`: in a nontrivial ring, the four designated product equalities cannot all hold.

Claims 1–3 need only a commutative ring; claims 4, 5 and 7 add nontriviality. Claim 6 uses `ZMod 1`. The explicit four-point control avoids mistaking global nonagreement for a finite-point statement.

Current source and limits

The current crypto main at `73d046ea68df459af9fa02c132cb0172024aa73d` retains the exact 1,035-byte `AffineTransition` definition/evaluation block from producer revision `8ce2800fbe3d265027e2341ec193ad69b946a8bd`: SHA-256 `ec1035ff7980da46be06338239144e62a27885623a8d20b23a79b426f6060bcc`. Other parts of `stark_v2.rs` changed. The proof does not certify Rust BigUint arithmetic, modular reduction, serialization, AIR fingerprint injectivity, Fiat–Shamir binding, FRI soundness, zero knowledge, or end-to-end security. The frozen target and scientific ledger are not changed. No new measurement or dashboard value follows from this composition.

The complete source-linked evidence table, reviewer SHA-256 values, receipt hashes and remaining obligations are in the sibling `report.md`.